

Coding and Smart Systems Outreach Program

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Speed Detector

Hardware Components Required:

- Raspberry Pi 4 with GPIO extension
- Breadboard
- One Infrared (IR) sensor
- One Active Buzzer
- One 220 Ω resistor
- Wires: red, black, purple, pink, yellow colors

Step-by-step instructions to connect hardware components:

1. Plug the sensor onto the breadboard.
2. Use a wire to connect **sensor VCC pin row to GPIO 5V** (pin 2).
3. Use another wire to connect **sensor GND pin to GPIO GND** (pin 6).
4. Use another wire to connect **sensor OUT pin row to GPIO 5** (pin 29).
5. Connect the buzzer **(Already done in motion sensor project)**
 - a. Place the buzzer on the breadboard.
 - b. Connect the **positive (+)** leg of the buzzer to **GPIO 27 (pin 13)** using a **yellow wire**.
 - c. Connect the **negative (-)** leg of the buzzer to **GND (pin 6)** using a **black wire**.
6. Power on Pi.
7. On the screen, **click on the projects folder**.
8. Click on the **speed_detector.py** file.
9. Write the code.

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Python code (Solution):

```
1 from gpiozero import DigitalInputDevice, Buzzer
2 from time import sleep, perf_counter
3
4 IR_PIN = 5
5 BUZZER_PIN = 17
6 CAR_CM = 12 # length of your car in cm
7
8 sensor = DigitalInputDevice(IR_PIN, pull_up=False)
9 buzzer = Buzzer(BUZZER_PIN)
10
11 try:
12     while True:
13         if not sensor.value: # car enters sensor
14             print("Car detected...")
15             start = perf_counter()
16             # wait until car leaves
17             while not sensor.value:
18                 sleep(0.001)
19             end = perf_counter()
20
21             elapsed = end - start
22             if elapsed < 0.01:
23                 print("Ignored unrealistically short reading")
24                 continue
25
26             speed_m_s = (CAR_CM / 100.0) / elapsed
27             speed_mph = speed_m_s * 2.23694
28
29             print(f"Elapsed: {elapsed:.4f} s ")
30             print(f"Speed: {speed_m_s:.2f} m/s ({speed_mph:.1f} mph)")
31
32             if speed_mph > 10.0:
33                 print("Speed over limit! Buzzing alert...")
34                 buzzer.on()
35                 sleep(1)
36                 buzzer.off()
37             sleep(0.01)
38
39 except KeyboardInterrupt:
40     buzzer.off()
```